

Collaborative FMCW Radar Swarm with Onboard AI Decision Engine for Aerial Surveillance

Aran Technologies | aranrobotics@gmail.com | IAF MBC-3 Competition

Abstract

We present a five-drone swarm as a distributed airborne FMCW radar -- a micro-AWACS architecture -- for real-time target detection, tracking, and ASP generation. Each drone carries six AWR1843 24 GHz panels (360-deg coverage) and processes detections through CFAR, Random Forest, and LLM tactical engine. First onboard ML+LLM inference chain on an airborne radar platform [1]; all behaviours verified in simulation.

1. Technical Approach

1.1 Platform. Five hexacopters (AUW 4.267 kg): six AWR1843BOOST AERIS-10 panels (24 GHz, 60-deg H-FOV, 360-deg combined), indigenous STM32 FC (custom PCB, FreeRTOS, 250 Hz PID, Kalman IMU), Doodle Labs AES-128 radio. Leader: Jetson AGX Orin 64 GB; soldiers: Orin NX 16 GB. Endurance ~32 min (6S 10,000 mAh). EKF2: optical flow + IMU + baro.

1.2 AI Pipeline. L1: AWR1843 DSP -- range-Doppler FFT + CFAR in <10 ms (range, azimuth, velocity, SNR). L2: Random Forest on Jetson -- target vs. clutter in <50 ms. L3: Llama 3.2 3B (leader) / Gemma 2B (soldiers) -- JSON situation reports to track reassignment, sector reorientation, threat alerts; triggers on L2 confirmations only.

1.3 Swarm Resilience. GCS > Leader > Soldier LLM. Leader loss >2 s: highest-battery soldier self-elects, resumes ASP fusion (sim. verified). Soldier loss: LLM redistributes orphaned tracks. ASP continuity >=3 drones (344 tracks logged). Split-merge on GCS command; re-fusion <5 s (sim. verified). Flask GCS 2.5 Hz: track table, sector map, decision log, JSON recording.

2. Innovation and Indigenisation

Novel: (i) first airborne CFAR->ML->LLM chain; (ii) LLM track reallocation on node failure [2,3]; (iii) priority-gated fallback without GCS. Indigenous: AI stack, ROS2, GCS, antenna PCB, airframe, STM32 FC. Contributes to AERIS-10 (github.com/NawfalMotii79/PLFM_RADAR) -- open-source 10.5 GHz pulse-LFM radar. COTS: AWR1843, Jetson, Doodle Labs. Indigenous >=55% (MBC-3 sec. 2.25). Nirmaan pre-incubation Phase 1, IIT Madras.

3. MBC-3 Compliance Summary

Requirement	Implementation
360-deg FOV / Range	6 x AWR1843 @ 60-deg intervals; 0.5-5 km; 120 m resolution
Multi-target / Revisit	10+ tracks; 72 raw slots/drone; revisit <1 s (sim. verified)
Graceful degradation	ASP maintained >=3 drones; bully election <2 s (sim. verified)
Endurance / AUW	~32 min, 6S 10000 mAh, 4.267 kg; op. height 500 m AGL min
Indigenisation	>=55% by mission-criticality weight (MBC-3 sec. 2.25)
GNSS-denied / RTH	EKF2: optical flow + IMU 250 Hz + baro; auto-RTH on link loss
Encrypted link / ASP	Doodle Labs AES-128; Flask GCS 2.5 Hz + JSON logs

4. Phase I Status

Phase I complete: SITL simulation verified (ASP, LLM reallocation, 344 tracks, split-merge), broadband and drone-to-ground comms tested, 4-min ISR demo video ready (github.com/lekkalaharsha/aran-mbc3-swarm). Phase II-III: hardware prototype, field demonstration.

References

- [1] Anon., "A Swarm of Drones for Detection and Localization of Airborne Targets," IEEE IGARSS, 2025.
- [2] X. Zhang et al., "Multimodal Large Language Models-Enabled UAV Swarm," arXiv:2506.12710, 2025.
- [3] PMC, "Multi-Agent Systems Powered by Large Language Models: Applications in Swarm Intelligence," Frontiers in AI, 2025.